MATS ToC

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Abstract

Novel pathogen emergencies compress scientific timelines. Early counter-breakout decisions must be made from sparse genomic signals while mitigating data scarcity and lack of expert input. This project proposes an emergency AI co-scientist for rapid macrocyclic-peptide countermeasure discovery. The system leverages existing tools for sequence annotation, gene-function prediction, and pathogen-family classification, and adds a conservative orchestration layer that produces evidence-linked target dossiers. On the design side, the co-scientist will propose macrocyclic peptide binders and generate concrete candidate configurations as inputs for experimental validation in peptide discovery workflows. We will evaluate the co-scientist end-to-end in a realistic toolchain using SynX's Discoverer platform, combining in silico stress tests and retrospective episodes with a gated wet-lab checkpoint that tests whether simulated winners advance across enrichment, binding, inhibition, and antiviral-activity stages. By measurably accelerating defensible therapeutic discovery from minimal inputs, this work aims to increase defensive readiness against both natural outbreaks and BAIM-enabled deliberate pathogen engineering, while setting clearer evaluation norms for agentic decision support in emergency biology.

Theory of Change

Emergencies compress scientific timelines: in a fast-moving pathogen outbreak, decision-makers must triage what to investigate and what to build under severe uncertainty and scarce experimental capacity. A response may begin with little more than a newly observed pathogen sequence from metagenomic surveillance in wastewater or other trasnferrable mediums, and the early hours and days are extremely decisive for narrowing to defensible targets and therapeutic leads. Because novelty is inherently unpredictable, we should not assume that the right domain experts are immediately available. A core uplift an AI co-scientist can provide is therefore speed and synthesis. By rapidly turning sparse early signals into coherent, evidence-linked can plans, it would significantly lower the minimum human-expertise threshold required to start a defensive therapeutic discovery program.

We believe macrocyclic peptides can act as the primary therapeutic discovery substrate due to their unique middle ground nature between traditional small molecules and antibodies. [1] Small molecules are easy to manufacture and can access intracellular biology, but they typically require deep, well-defined binding pockets and therefore leave many targets (especially protein–protein interfaces) effectively "undruggable." Antibodies can bind large, flat protein surfaces with high specificity, but they are generally limited to extracellular targets and often have poor access to intracellular compartments and some dense tissues. Macrocyclic peptides can present a larger binding surface than small molecules while remaining substantially smaller than antibodies, enabling engagement of challenging interfaces and, with appropriate design, increasingly supporting intracellular targeting and membrane diffusion. [1] [2] Predicting cyclic-peptide permeability is now a benchmarked ML subproblem, which makes progress measurable rather than anecdotal. [3]

Macrocyclic peptide design alone is not sufficient for excellent emergency response. The speed of synthesizing partial evidence into the next best action is a primary limiting factor. The core theory of change is that many high-impact failures in fast-moving technical domains are mediated by bad decisions under uncertainty rather than a lack of any single capability. Improving decision quality requires explicit uncertainty tracking, conservative updating as genetic evidence arrives, and resistance to brittle reasoning under epistemic distribution shift. [4]

Herein, this MATS project formalizes an emergency AI co-scientist that turns minimal inputs, such a poorly characterized pathogen gene sequence, into ranked, evidence-backed macrocyclic drug targets. We place an emphasis on hands-off agentic orchestration. Wherever possible, the co-scientist autonomously leverages best-available existing tools for sequence annotation, gene-function prediction, and pathogen-family classification, and adds a conservative layer that traces claims to evidence, proposing next actions under capacity constraints. Regarding peptide design, the co-scientist will propose actionable targets and then design macrocyclic peptide binders and candidate sets intended for experimental validation in peptide discovery workflows. [5] [6]

Testbed experiments will be conducted using SynX's Discoverer design platform. A full experimental loop connecting sequence-level emergency briefs and target-and-assumption dossier to rapid proposal of peptide family designs will be deployed. The evaluation plan includes both in silico components and a wet-lab validation checkpoint that gates the claims of the co-scientist. [7] [8] For structure-based scoring and interface design with noncanonical amino acids, we can leverage established Rosetta approaches and tooling. [9] [10] [11] Recent deep-learning methods also suggest that high-affinity protein-binding macrocycles can be designed with increasing reliability. [12] More general atom-level binder models provide an additional reference point for the direction of travel in de novo binder design. [13] [14]

Impact and reducing catastrophic risk from AI

This work aims to mitigate risk from increasingly capable Biological Artifical Intelligence Models (BAIMs), which lower barriers to engineering novel non-natural pathogens, enabling malicious actors to cause large-scale loss of life. This project reduces that risk indirectly by improving emergency countermeasure readiness. By aiming to make "therapeutic design under severe uncertainty" a more reliable and evaluable workflow,

biotech will be able to respond faster as a field when a novel pathogen, whether man-made or natural, begins to threaten public health and human life.

This project is intentionally narrower than "general" AI scientists (e.g., broad autonomous research agents from DeepMind and FutureHouse). From a defense-dominance perspective, we aim to achieve specific capability uplift for emergency macrocyclic-peptide discovery. The co-scientist should accelerate the path from early sequence evidence to testable peptide designs while lowering the level of specialized human expertise required to get started. To make this uplift measurable, the co-scientist will simultaneously create evaluation episodes and a failure taxonomy for agentic decision-making in peptide biology. [4].

Futures of Concern and Net-Negative Risks

The primary futures of concern involve (i) offensive misuse—actors using BAIM-enabled design to accelerate pathogen development—and (ii) premature deployment of tool-using AI agents in critical biomedical workflows before reliability standards catch up. A secondary issue is that evaluation practice remains focused on isolated correctness rather than end-to-end decision quality under constraints, allowing systems that look impressive in demos to fail in operational use.

A central technical risk is epistemic: historical evaluation (e.g., January–March 2020 COVID-19 evidence, which is a case study we will use to evaluate the co-scientist) can suffer from hindsight leakage, including knowledge embedded in model weights. Domain-specific issues, such as conflating endpoints (i.e. enrichment, binding, inhibition, and antiviral activity), overinterpreting binding-affinity estimates without a fixed protocol, and drawing conclusions from biased measurement datasets taken from emergency conditions, may also rise. [15]

For future works to mitigate these risks, the project may utilize a worst-case uplift biological evaluation harness constructed with Thinking Machines' Tinker platform. Evaluations of the AI co-scientist will validate signals related to enrichment, binding, inhibition, and antiviral activity, while penalizing overconfident recommendations that do not survive experimental contact. By doing so, we aim to reduce biological harms from miscalibrated agentic deployment and help shift norms toward decision-support agents that justify recommendations against a time-limited evidence base, quantify uncertainty, and behave stably under irrelevant updates while changing appropriately under relevant evidence.

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